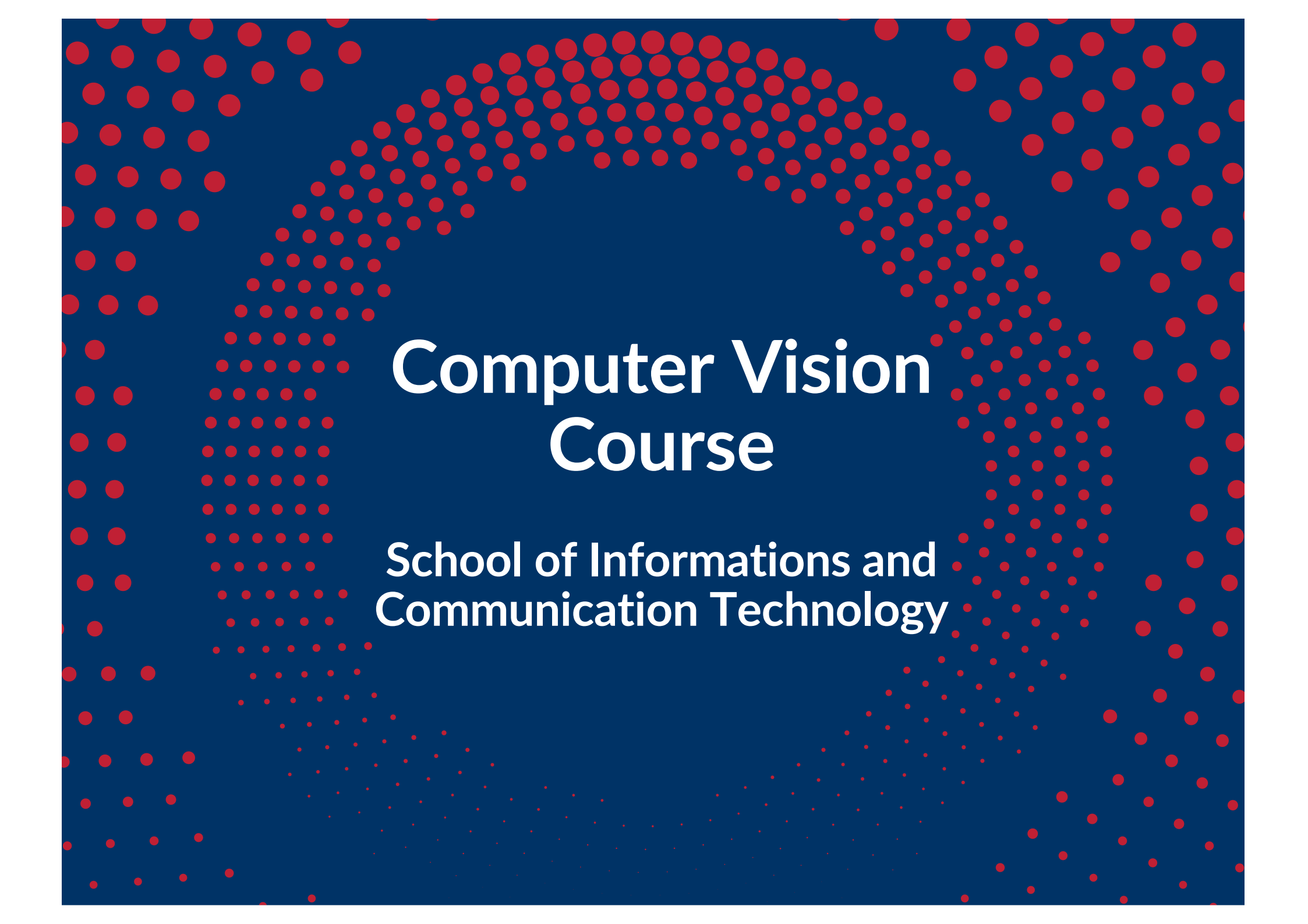




# Computer Vision Course

School of Informations and  
Communication Technology



ĐẠI HỌC  
BÁCH KHOA HÀ NỘI  
HANOI UNIVERSITY  
OF SCIENCE AND TECHNOLOGY

# Computer Vision

## Chapter 2: Image formation, acquisition and digitization

Computer Vision Group

School of Information and Communications Technology

ONE LOVE. ONE FUTURE.

- Image formation
  - Human vision
  - Image formation
- Acquisition and digitization: Digital camera
  - Imaging sensor
  - 2D signal and sampling
- Color:
  - Primary color, additive/ subtractive color, color spaces
- Digital image representation and formats

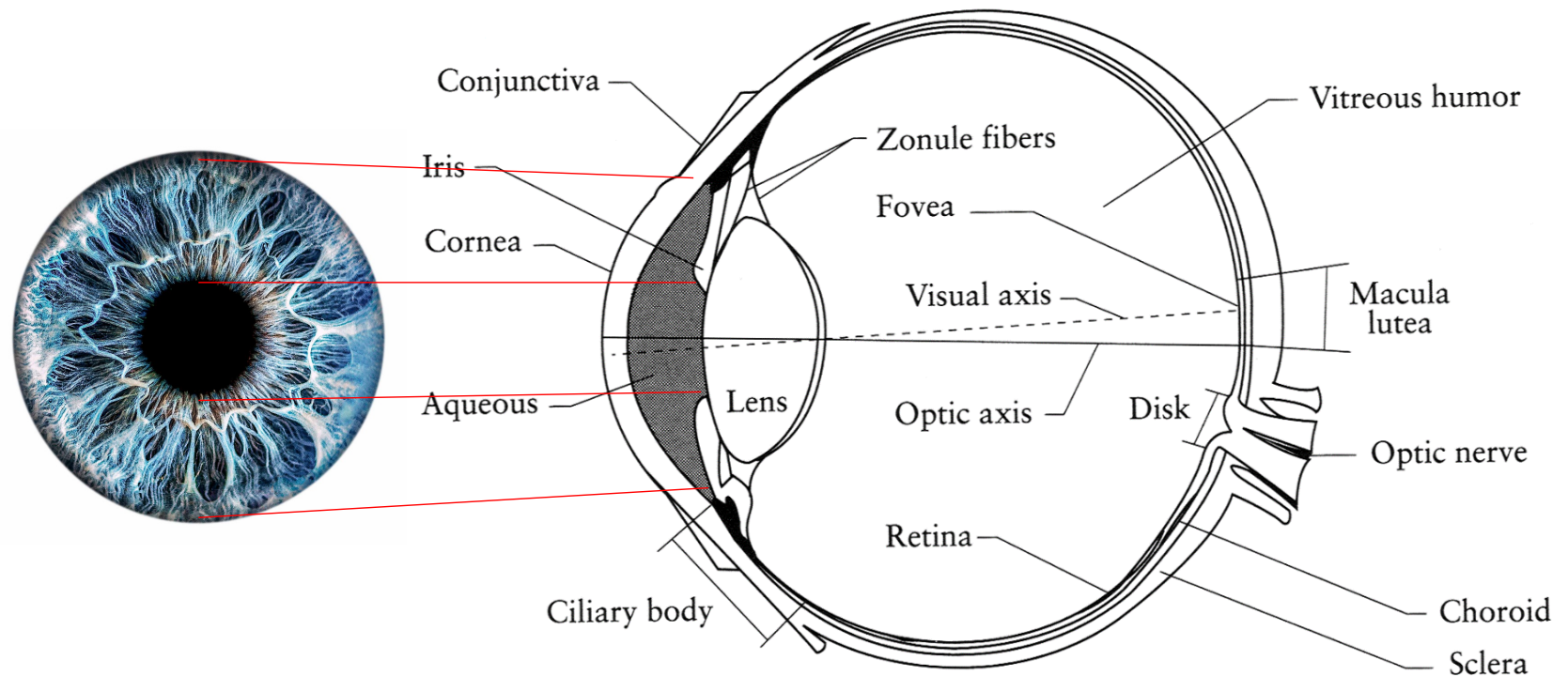
# Image formation

**Image formation** studies the forward process of producing images and videos.

- Image formation encompasses the radiometric and geometric processes by which 2D images of 3D objects are formed. To produce a real image, the nature of the visual sensors (i.e. CCD and CMOS cameras), should be studied.
- Imaging process is a [mapping of an object to an image plane](#).
- With [digital images](#), the image formation process also includes analog to digital conversion, [sampling](#)
- **Human color vision (Perception)** : In the case of computer vision the light incident on the sensor comprises the image. In the case of visual perception, the human eye has a color dependent response to light which is the spectral sensitivity of human vision.

# The Eye

- The human eye is a camera
  - **Iris** - colored annulus with radial muscles
  - **Pupil** - the hole (aperture) whose size is controlled by the iris
  - What's the sensor?



– photoreceptor cells (rods and cones) in the **retina**

Slide by Steve Seitz

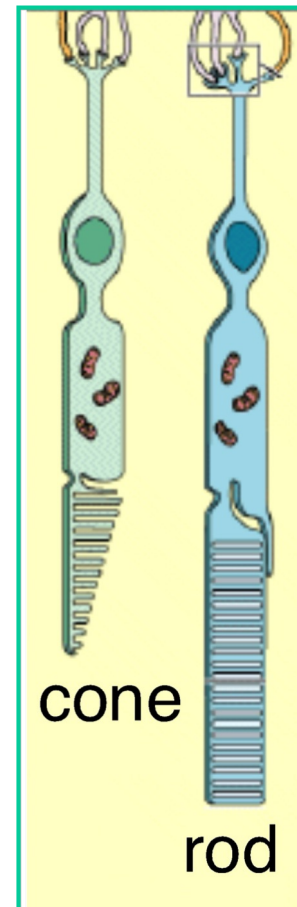
# Two types of light-sensitive receptors

## **Cones**

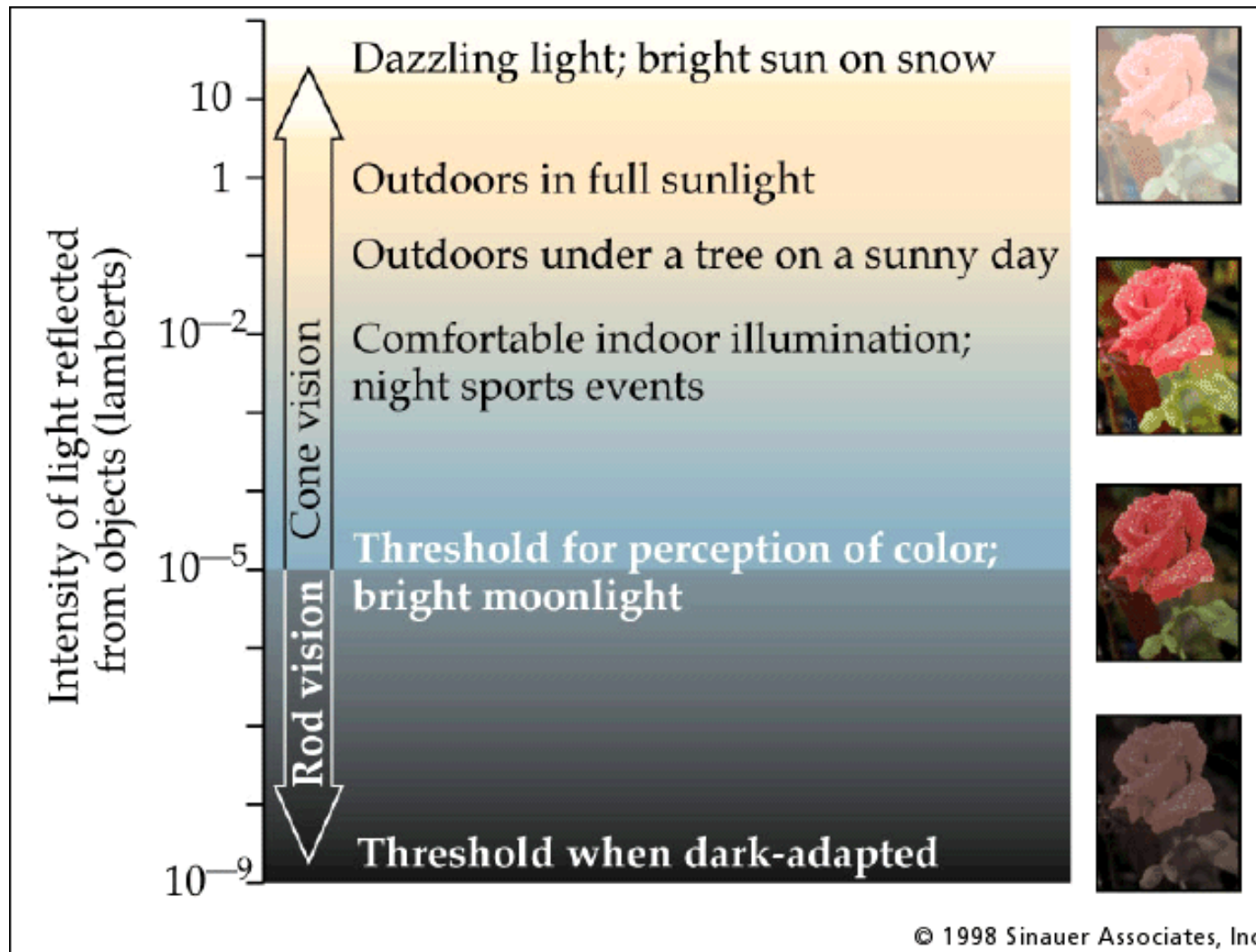
cone-shaped  
less sensitive  
operate in high light  
color vision

## **Rods**

rod-shaped  
highly sensitive  
operate at night  
gray-scale vision

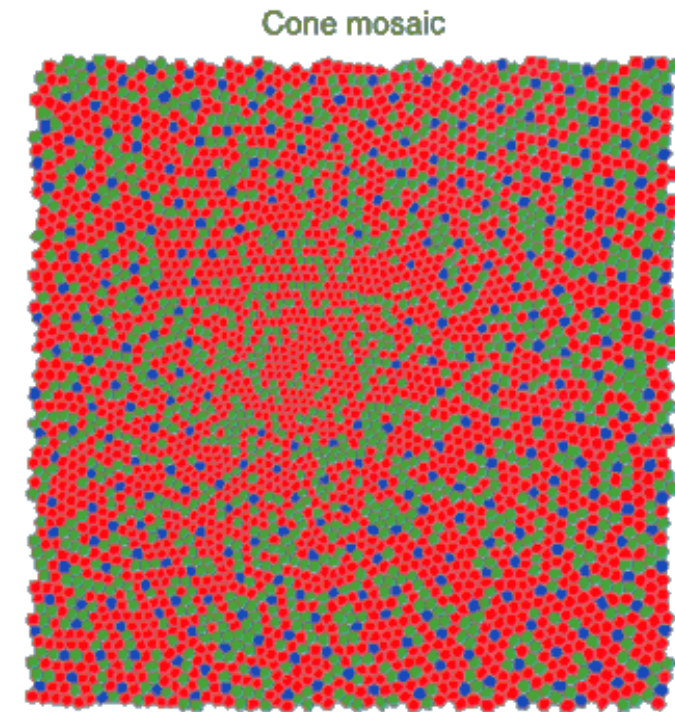
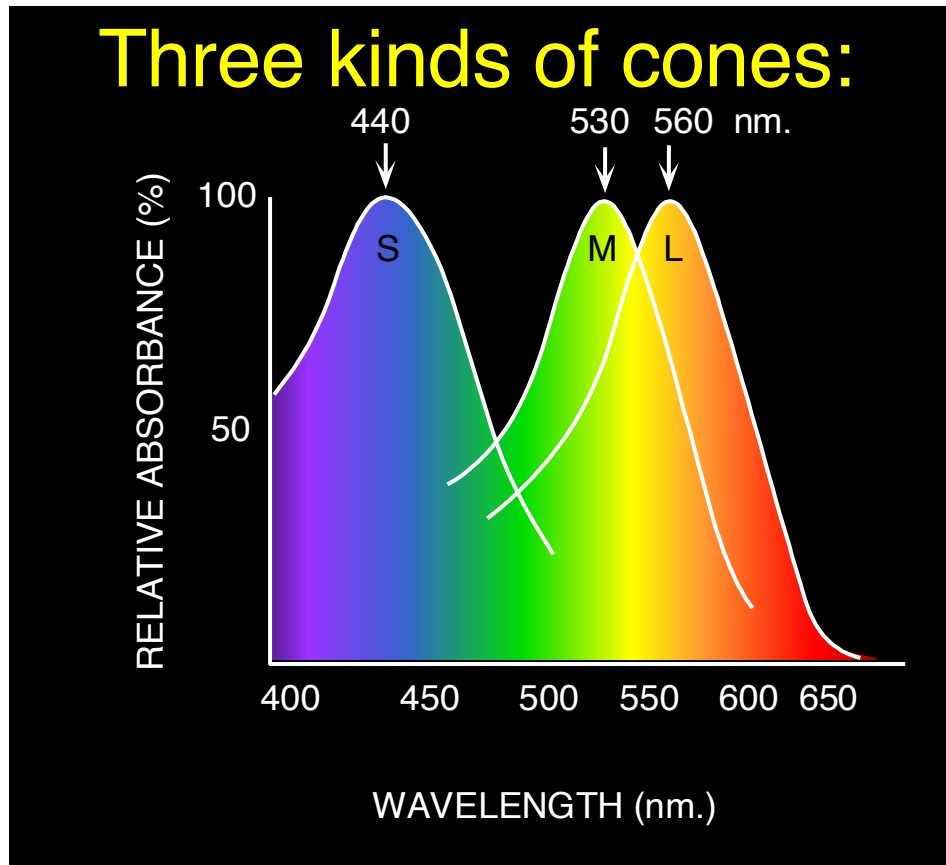


# Rod / Cone sensitivity





# Physiology of Color Vision

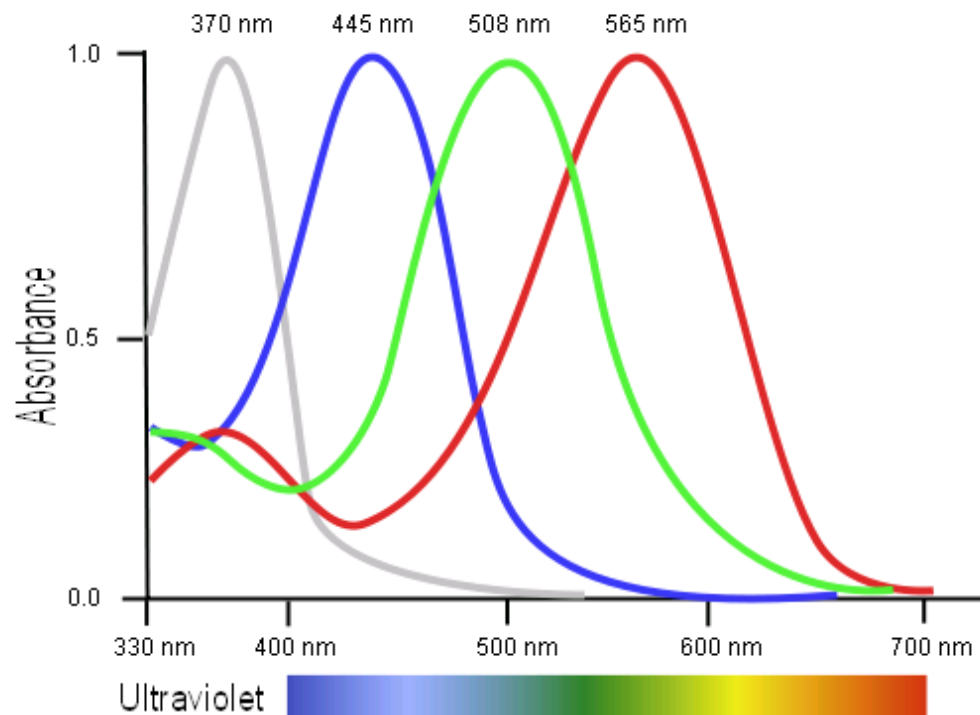


© Stephen E. Palmer, 2002



# Tetrachromatism

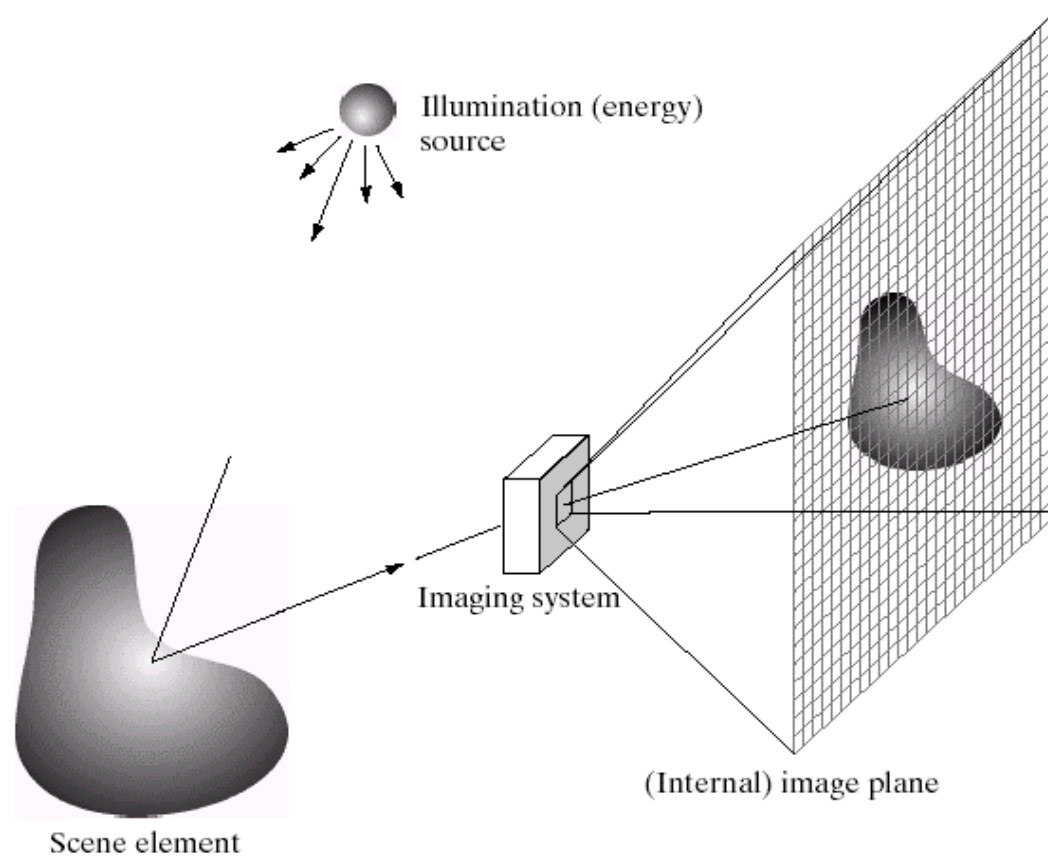
- Most birds, and many other animals, have cones for ultraviolet light.
- Some humans seem to have four cones (12% of females).



Bird cone  
responses

James Hays

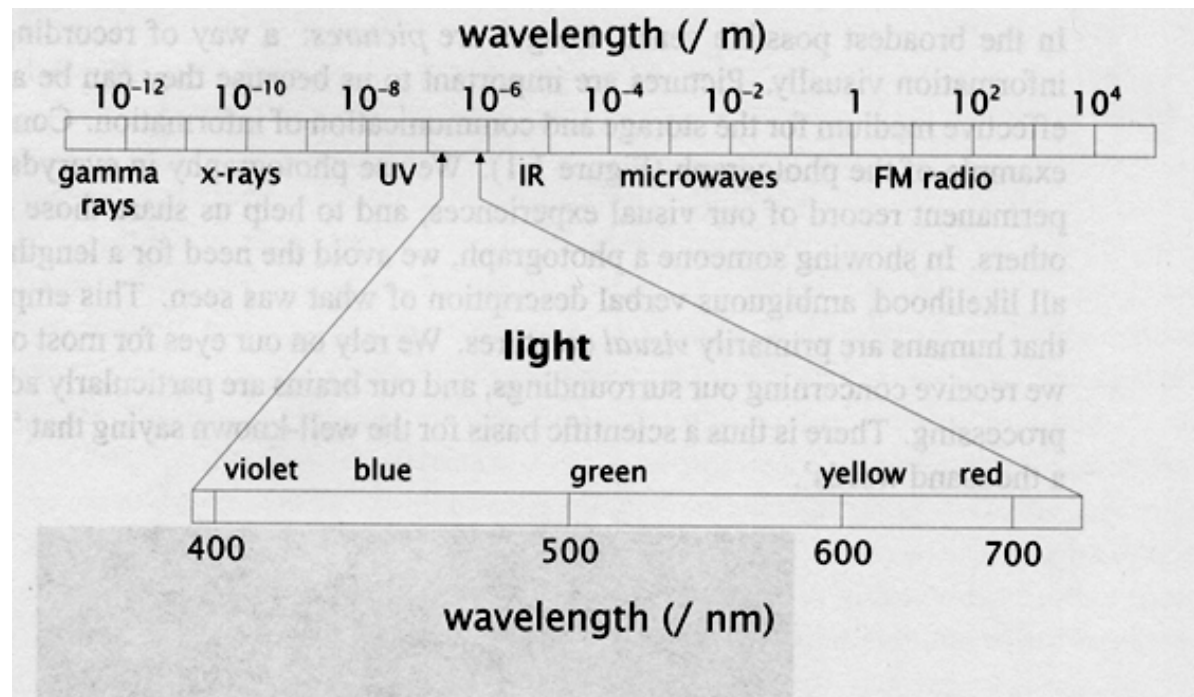
# Image formation



Adapted from S. Seitz

# What is light?

- Light: The visible portion of the electromagnetic (EM) spectrum.
- Light occurs between wavelengths of approximately 400 and 700 nanometers.



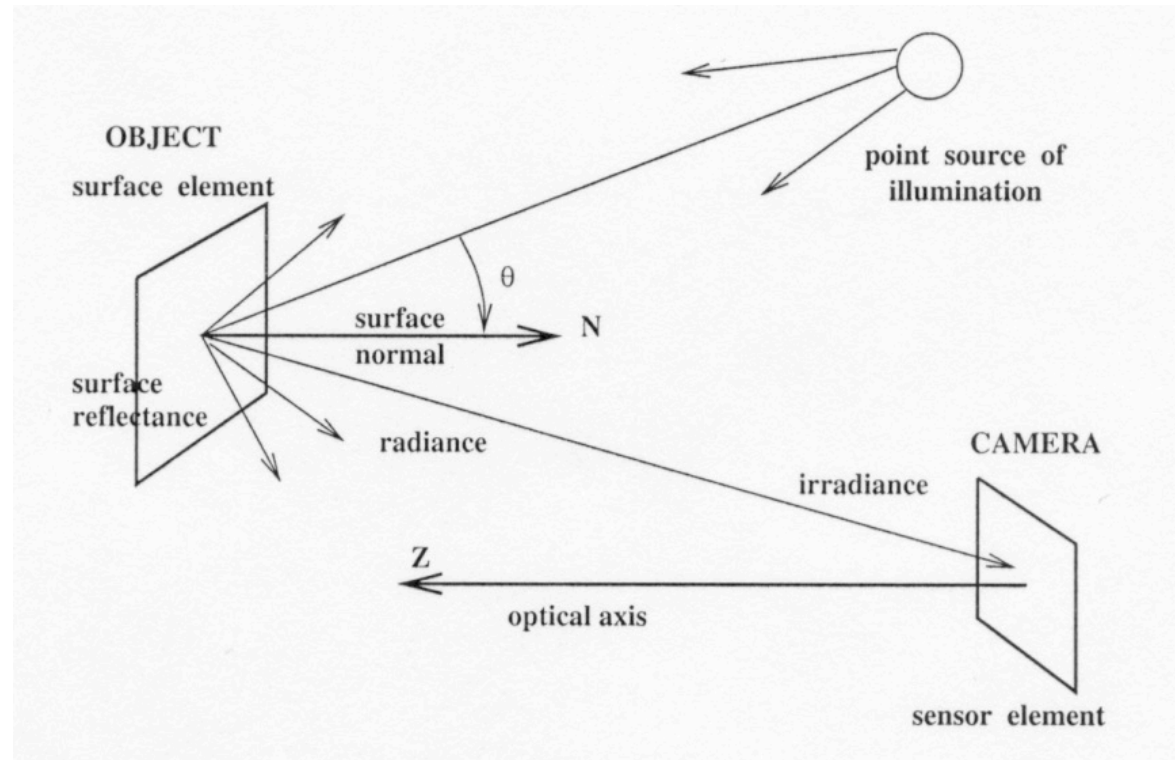
# Photometric image formation

- Illumination source: Sun, light ...
- Photometric measurement:
  - Perceptual brightness of visible electromagnetic energy of light.
- Optical system (lenses):
  - An object (scene) may be illuminated by the **light from an emitting source**.
  - The light incident on the object is **reflected in a manner dependent on the surface properties of the object**
  - An illuminated object will scatter **light toward a lens** and the **lens will collect and focus the light** to create the image
- Imaging sensor: **CCD (charge-coupled device)** or **CMOS** sensors cameras provide the 2D sensed signal.
- Digital camera: 2D sensed signal is pass to ***analog-to-digital converter*** (*sampling*), it create the digital image

# Photometric image formation

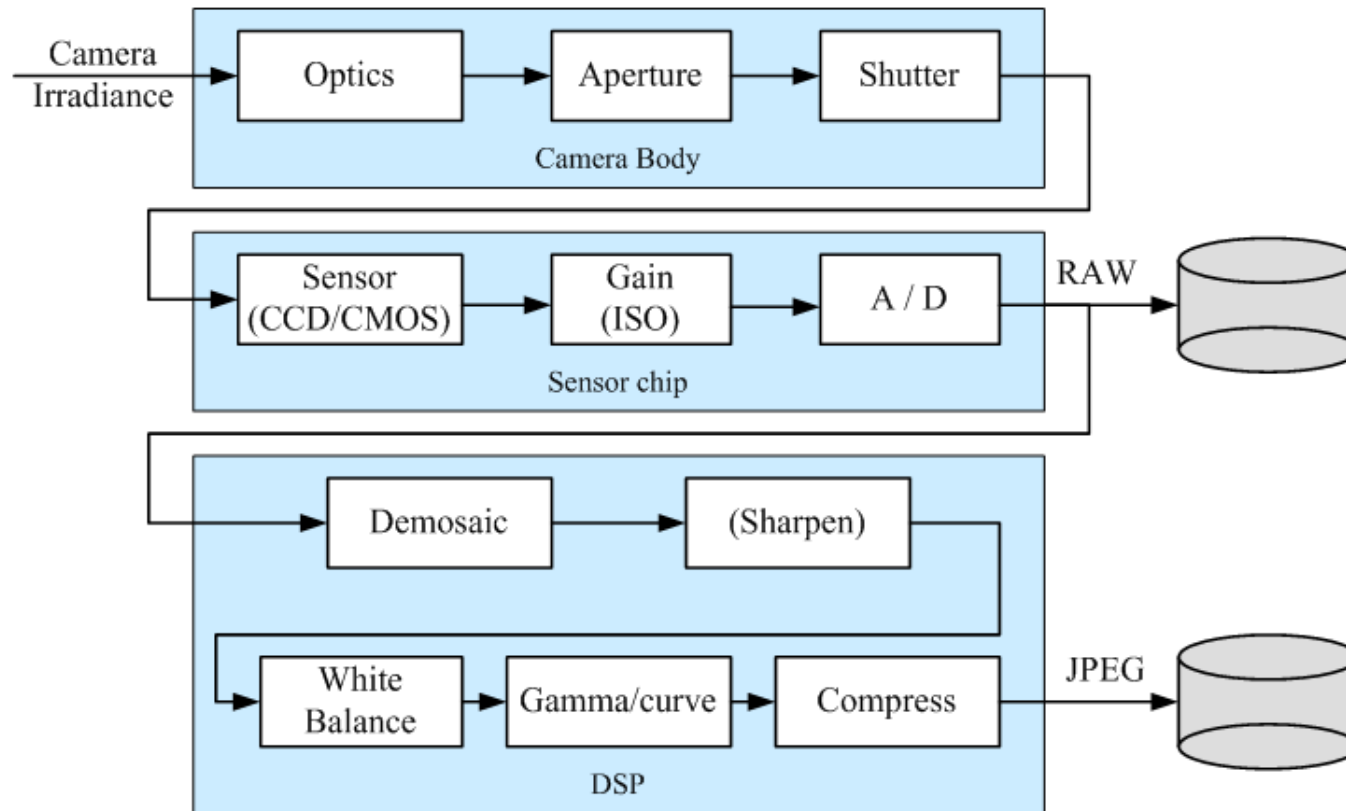
- Modeling the image formation process: 3D geometric features in the world are projected into 2D features in an image.
- A simplified model of photometric image formation is illustrated.

- The scene is illuminated by a single source.
- The scene reflects radiation towards the camera.
- The camera senses it via CCD/ CMOS



# Acquisition and digitization: Digital camera

## Digital camera: Image sensing and processing pipeline

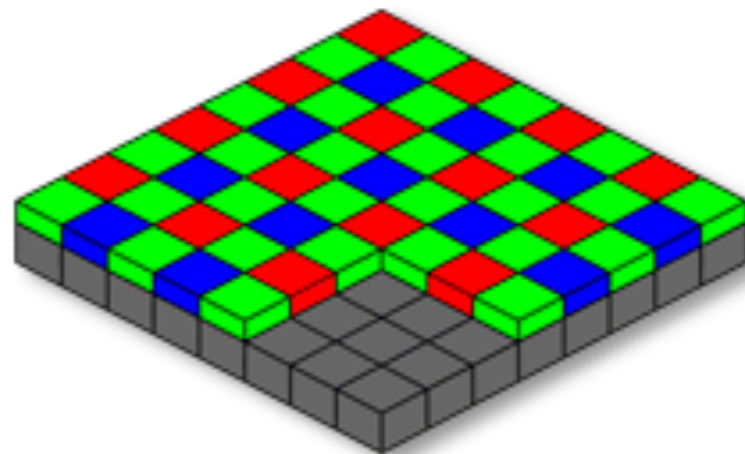


*Adapted from S. Seitz*

- **Image acquisition:**
  - Optical system, aperture (capture), shutter
  - Imaging sensor: CCD/ CMOS sensor camera consists of a array of photodiodes. Each cell in the is light-sensitive diode that converts photons to electrons.
  - 2D sensed signal of image, video
- **Digitization (ADC): Sampling and Quantization**
  - Sampling the 2D sensed signal create the samples or pixels
  - Quantizing the sample values as the integer values of pixels
- **Processing (DSP- Digital Signal Processing):**
  - Cameras perform a variety of digital signal processing operations *to enhance* the image before *compressing* and *storing* the pixel values in standard format file.

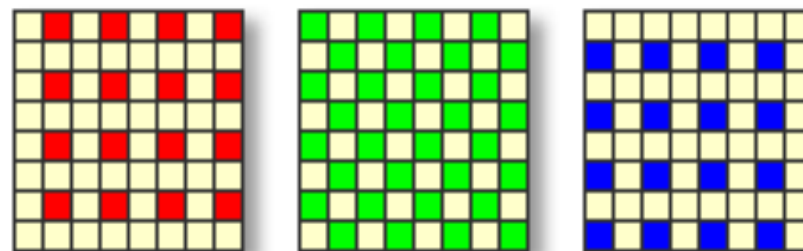
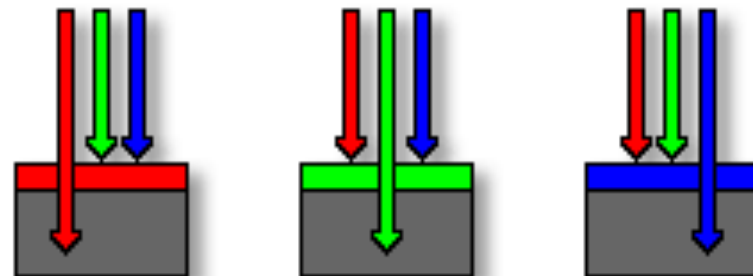


# Sensor array : an example

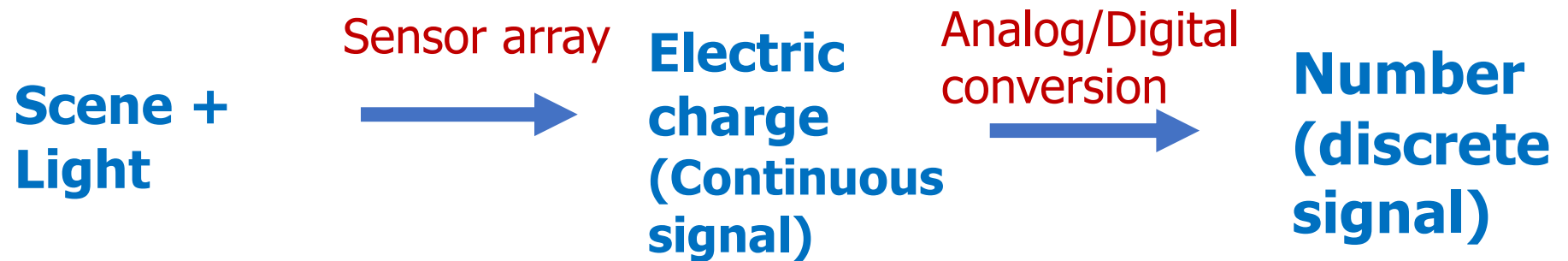


Capteur photosensible  
recouvert d'une grille de Bayer

Rayonnement



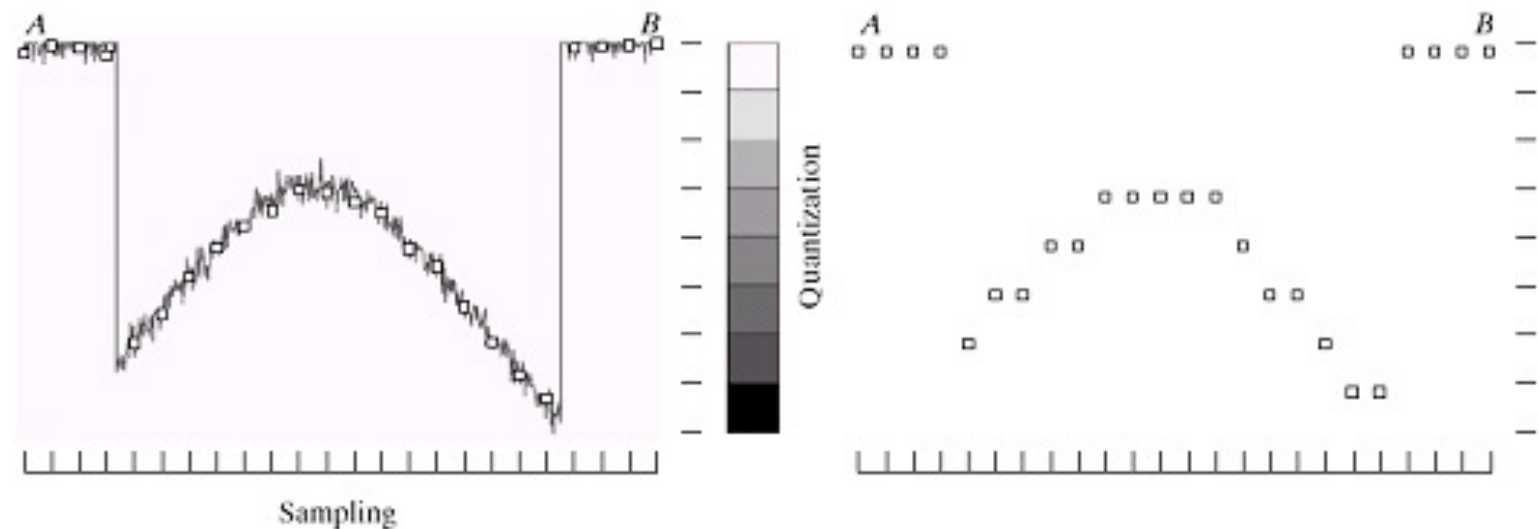
# Real scene -> digital Image



**Digitization = Sampling + Quantization**

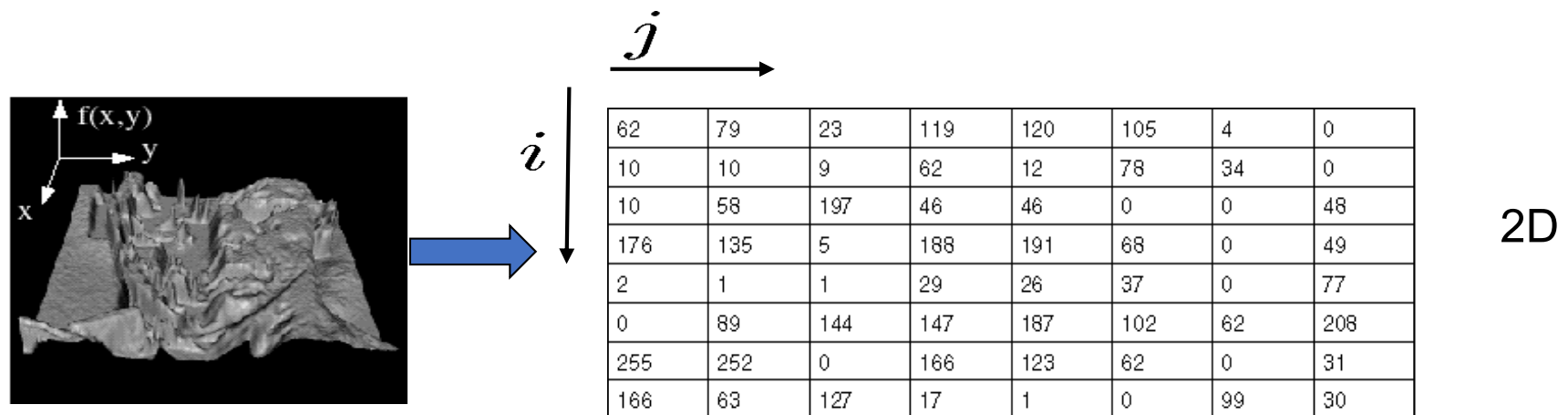
# Sampling and quantization

- **Sample** the 2D space on a regular grid
- **Quantize** each sample (round to nearest integer)

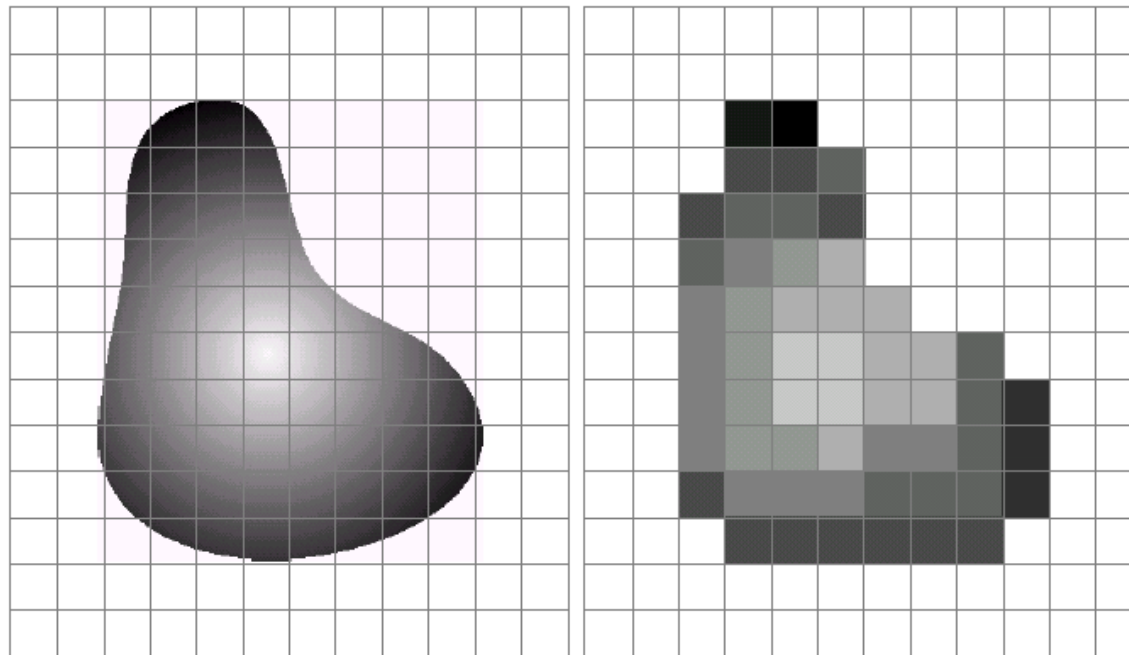


# Sampling and quantization

- **Sample** the 2D space on a regular grid
- **Quantize** each sample (round to nearest integer)



# Digital image



a b

**FIGURE 2.17** (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.

# Spatial resolution (sampling)



200 X 278



50 X 70



12 X 18

# Gray-level resolution (Quantization)



8 bits



4 bits



2 bits

# Color spaces

- Color spaces; different types of color modes
- Color represented by vector of components
  - ❖ Red, Green, Blue (**RGB**)
  - ❖ Hue, Saturation, Value (**HSV**)
  - ❖ Luminance, chrominance (**YUV**, LUV)
  - ❖ **XYZ**
- Color convert: RGB – YUV

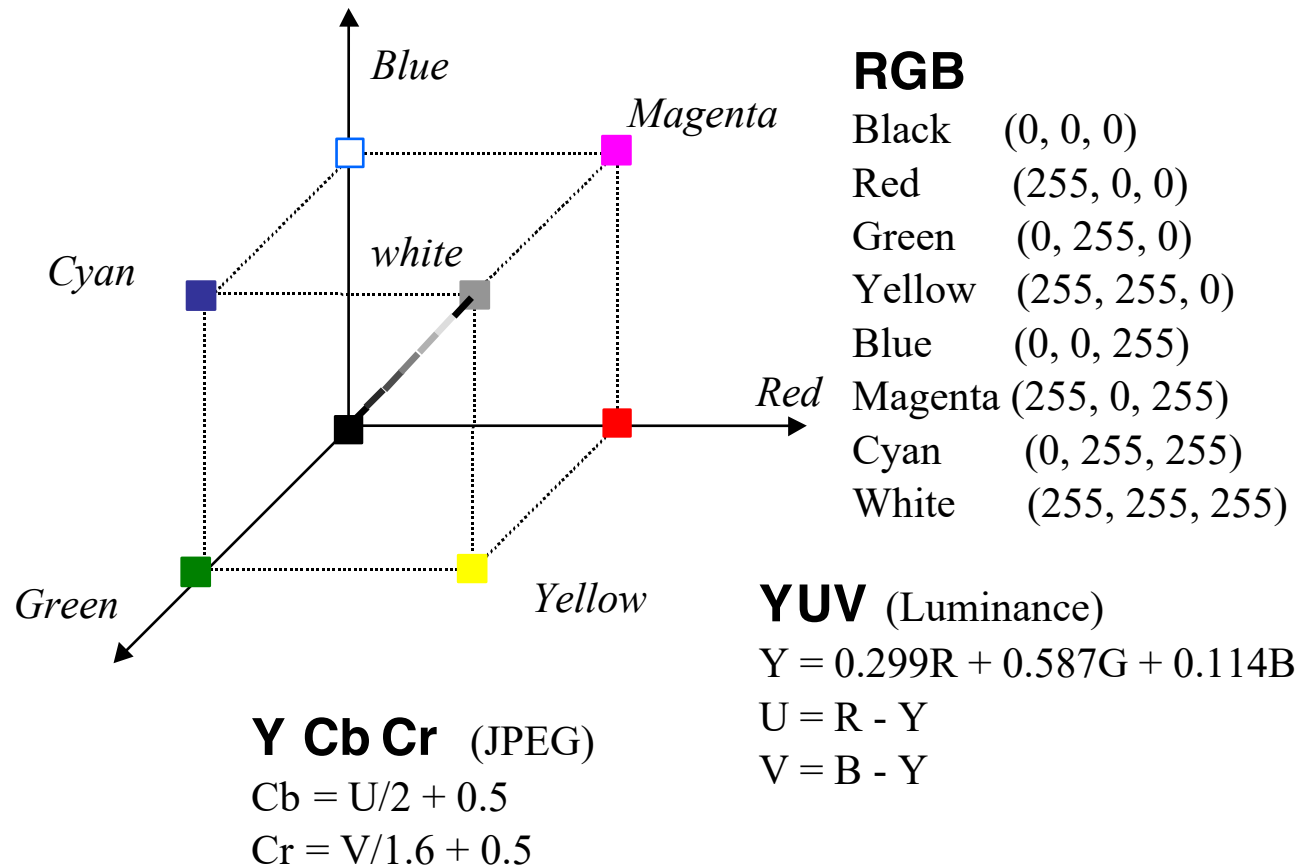
$$Y = 0.299R + 0.587G + 0.114B$$

$$U = 0.493 (B - Y) ; V = 0.877 (R - Y)$$

$$\begin{bmatrix} Y \\ C_R \\ C_B \end{bmatrix} = \begin{bmatrix} 0.257 & 0.504 & 0.098 \\ 0.439 & -0.368 & -0.071 \\ -0.148 & -0.291 & 0.439 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} + \begin{bmatrix} 16 \\ 128 \\ 128 \end{bmatrix}$$



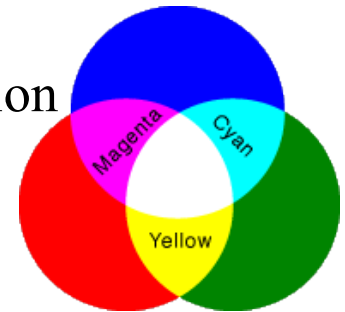
# Color coordinate system



# Color: Additive/Subtractive primary color

- **Primary color:** Red, Green, Blue (RGB)
- **Additive colors:**
  - Combination of RGB can be mixed to produce Cyan, Magenta, Yellow (CMY) & White.
  - *Additive color reproduction system:*
    - Combination of RGB to reproduce a colored light.
- **Subtractive colors** CMY can be mixed to produce RGB & black
  - *Subtractive color reproduction system:* A white light sequentially passes through cyan, magenta, yellow filters to reproduce a colored light.

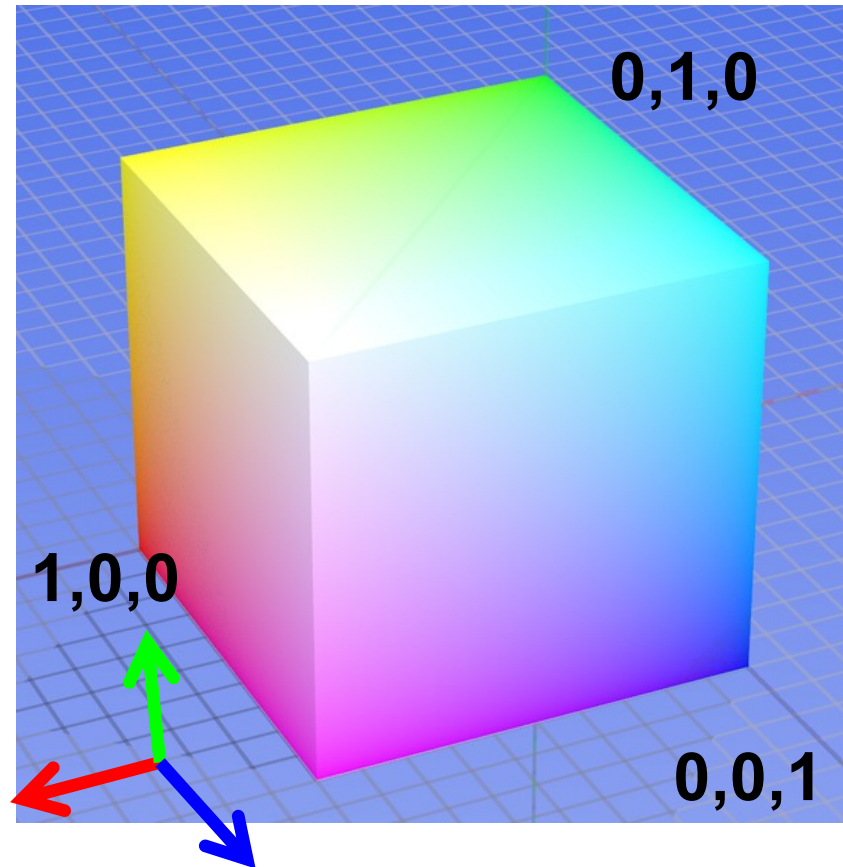
Colors:  
combination  
of RGB



# Color spaces: RGB



## Default color space



Any color =  $r \cdot R + g \cdot G + b \cdot B$

- Strongly correlated channels
- Non-perceptual



**R = 1**  
(G=0,B=0)



**G = 1**  
(R=0,B=0)

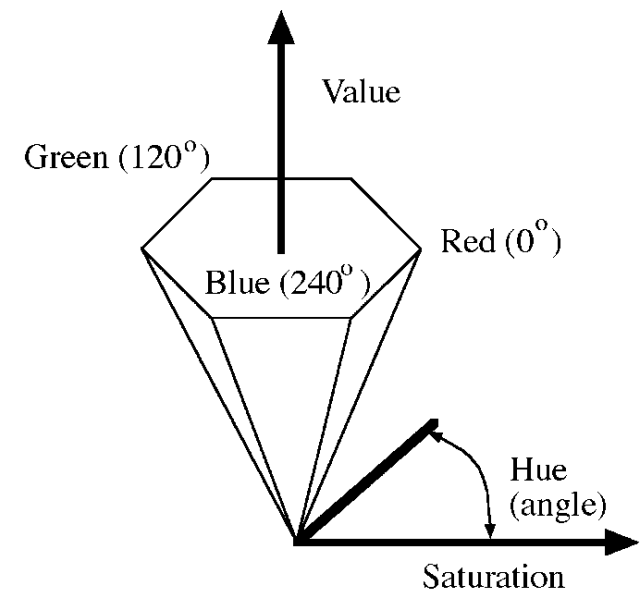
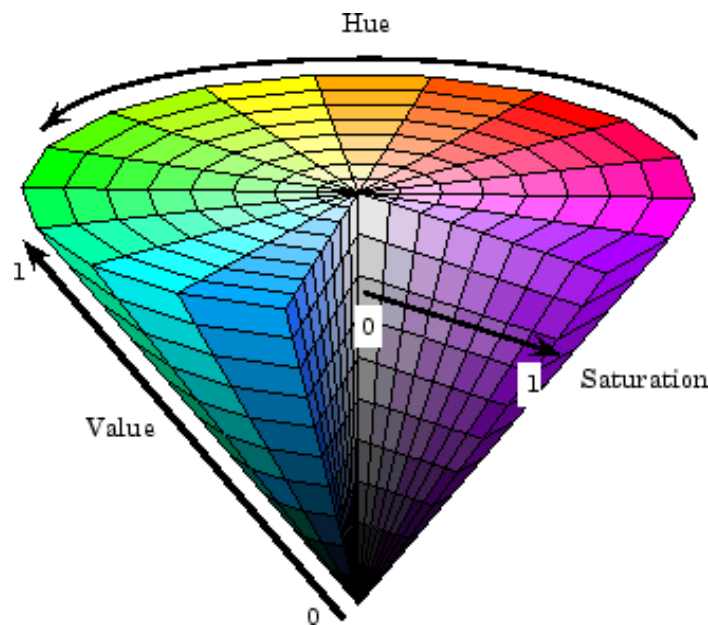
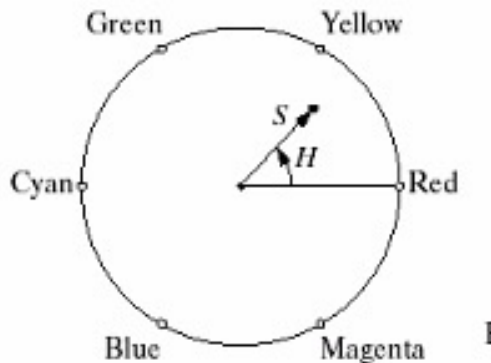
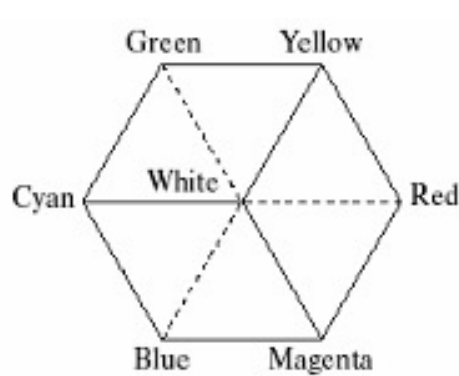


**B = 1**  
(R=0,G=0)

Image from: [http://en.wikipedia.org/wiki/File:RGB\\_color\\_solid\\_cube.png](http://en.wikipedia.org/wiki/File:RGB_color_solid_cube.png)

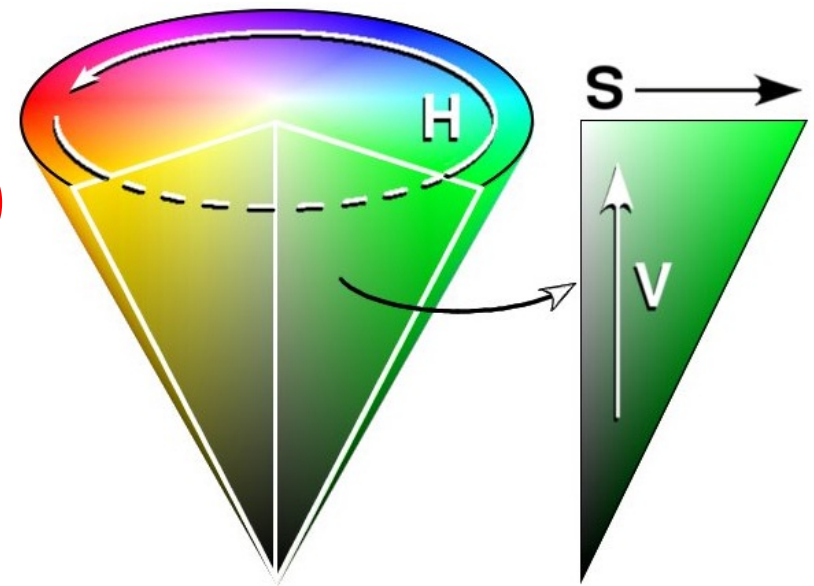
# Nonlinear color spaces: HSV

- Perceptually meaningful dimensions:
  - Hue, Saturation (chroma)
  - Value (Intensity)



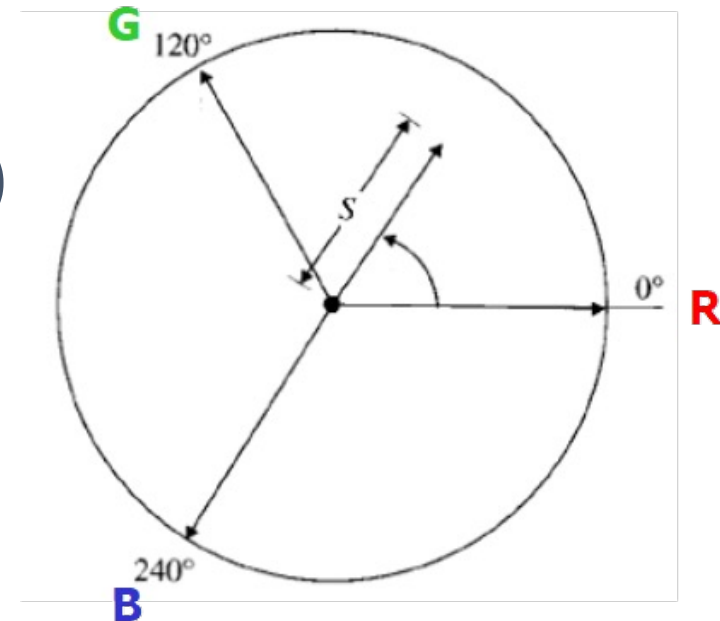
# HSV (Hue – Saturation- Value)

- The Hue-Saturation-Value (HSV) color space is useful for segmentation and recognition
  - Non-linear conversion
  - Visual representation of colors
- We identify for a pixel:
  - The pixel *intensity (value)*
  - The pixel *color (hue + saturation)*
- RGB does not have this separation



# HSV (Hue – Saturation- Value)

- Hue (H) is coded as an angle between 0 and 360
- Saturation (S) is coded as a radius between 0 and 1
  - $S = 0$  : gray
  - $S = 1$  : pure color
- Value (V) = MAX (Red, Green, Blue)



# HSV (Hue – Saturation- Value)

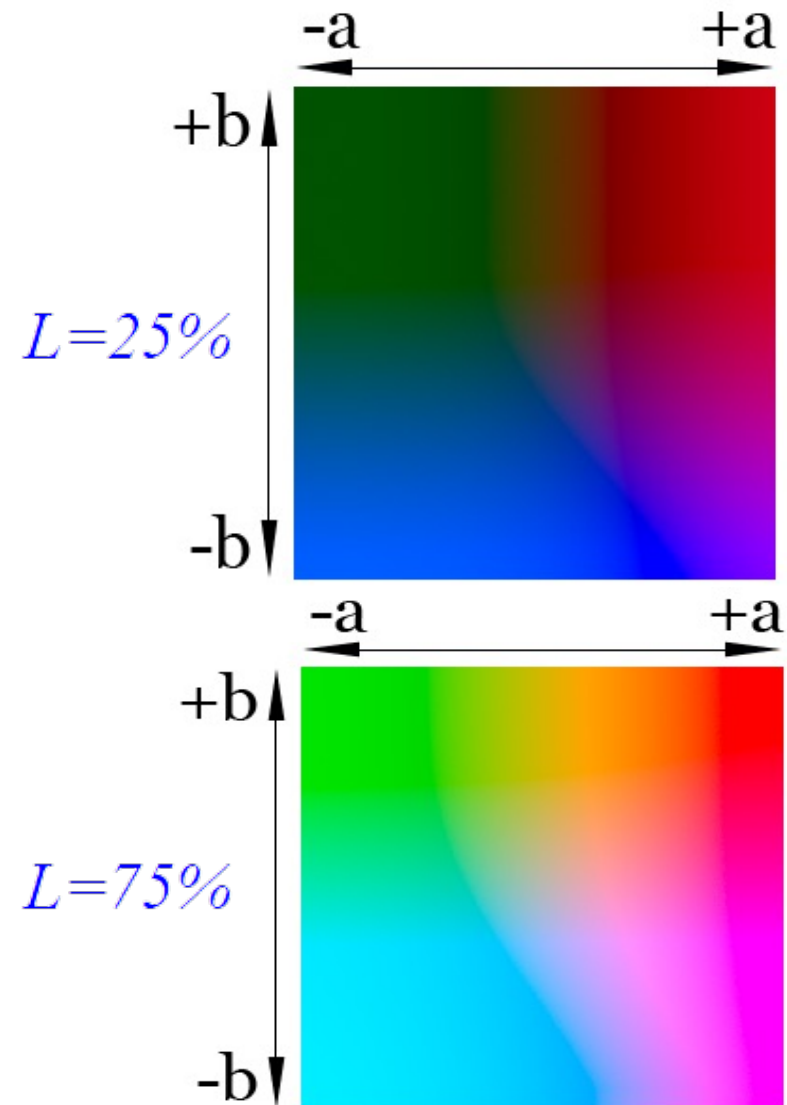
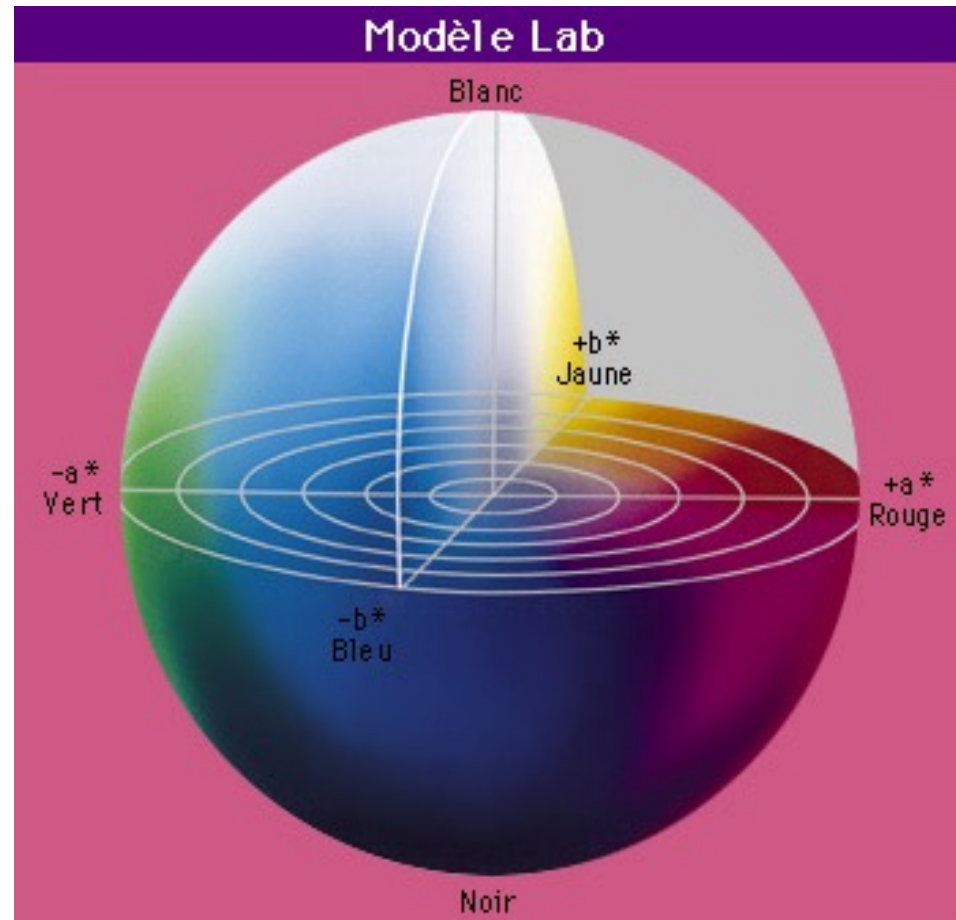
- If we know the color of the object we are looking for, we can model it using a **hue interval**
- Take care, because it is an angle (periodic value)
  - Hue < 60° means nothing
    - *Is 350° smaller or bigger than 60°?*
  - Define an interval: 350° < Hue < 60° (for example)
- This interval is valid **if Saturation > threshold** (otherwise gray level)
- This is **independant of Value**, which is more sensible to light conditions

# Lab color space

- The **Lab** system (sometimes  $L^*a^*b^*$ ) is based on a study of human vision
  - independant from all technologies
  - presenting colors as seen by the human eyes
- Colors are defined using 3 values
  - **L** is the luminance, going from 0% (black) to 100% (white)
  - $a^*$  represents an axis going from green (negative value, -127) to red (positive value, +127)
  - $b^*$  represents an axis going from blue (negative value, -127) to yellow (positive value, +127)

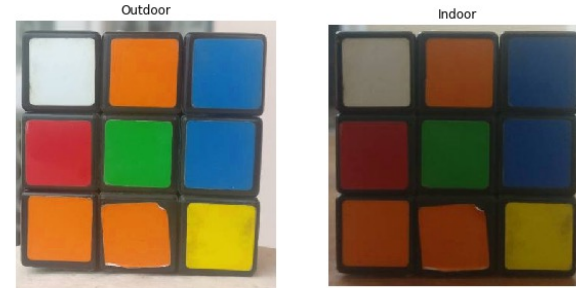


# Lab color space



# Color space vs. illumination conditions

- collected 10 images of the cube under varying illumination conditions



- separately cropped every color to get 6 datasets for the 6 different colors



Changes in color due to varying Illumination conditions

- Compute the density plot: Check the distribution of a particular color say, blue or yellow in different color spaces. The density plot or the 2D Histogram gives an idea about the variations in values for a given color

# Color space vs. illumination conditions

- Similar illumination: very compact

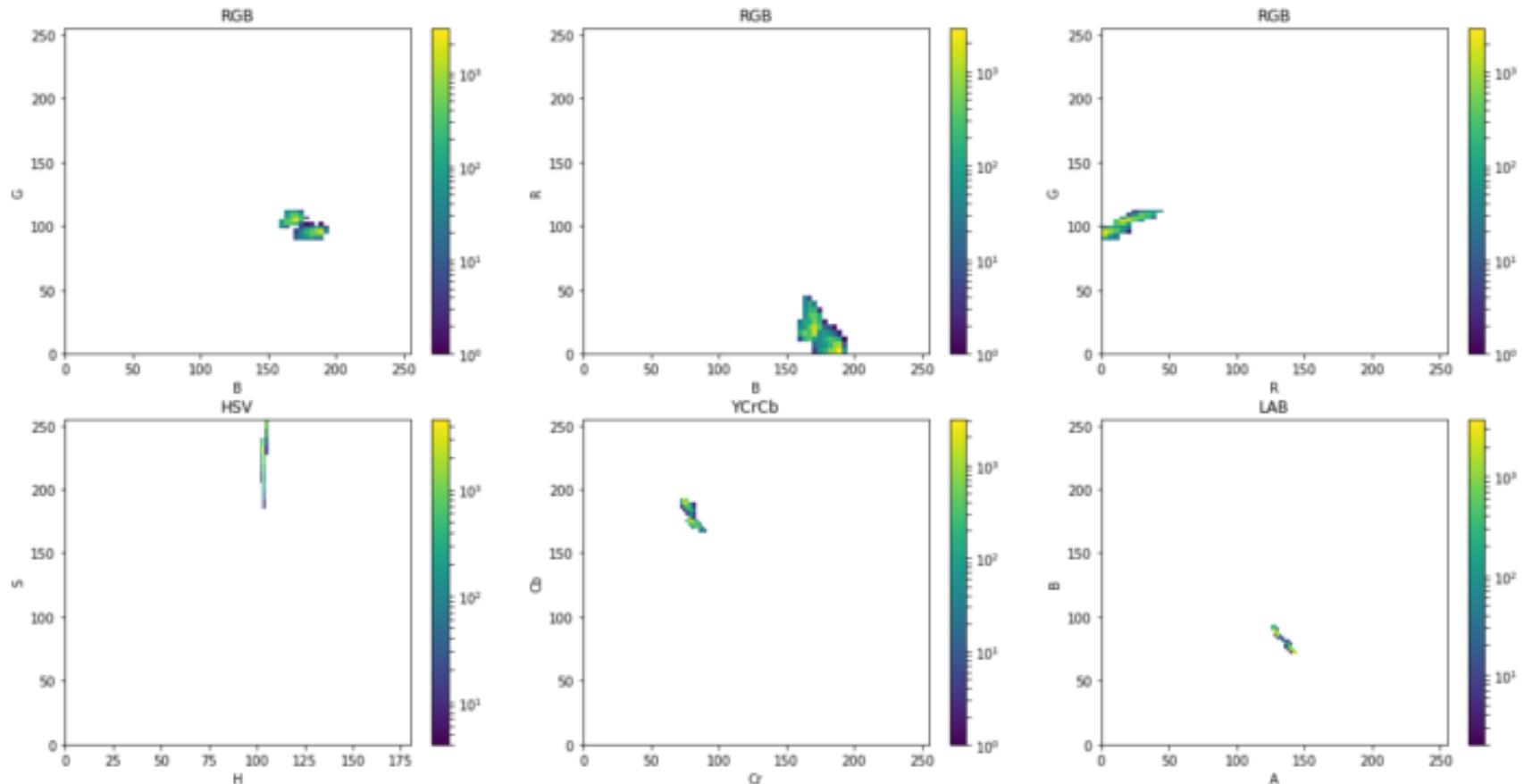


Fig.: Density Plot showing the variation of values in color channels for 2 similar bright images of **blue color**

Source: Vikas Gupta, Learn OpenCV

# Color space vs. illumination conditions

- Similar illumination: very compact

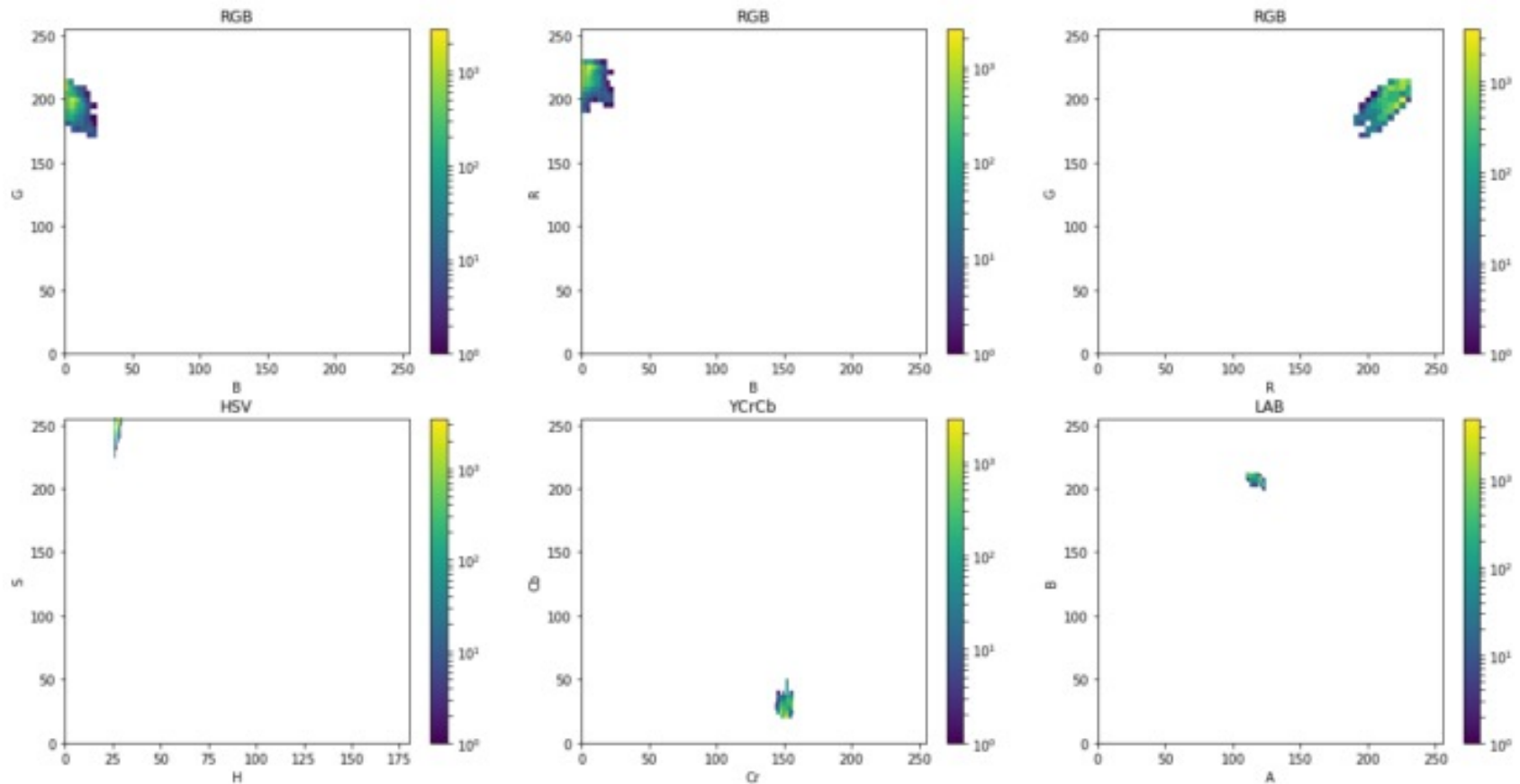


Fig.: Density Plot showing the variation of values in color channels for 2 similar bright images of **yellow color**

# Color space vs. illumination conditions

- Different illumination:

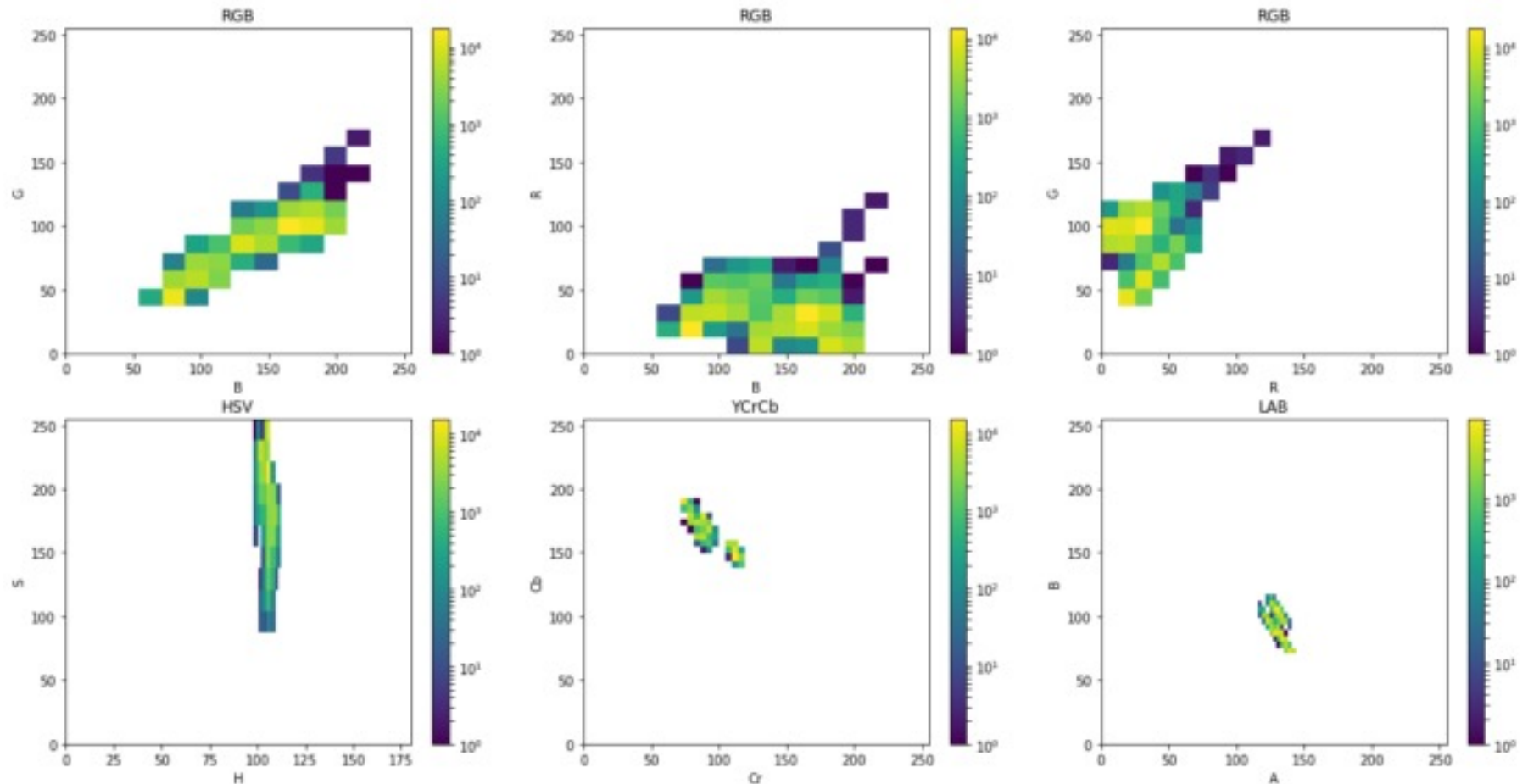


Fig.: Density Plot showing the variation of values in color channels under varying illumination for the **blue color**

Source: Vikas Gupta, Learn OpenCV

# Color space vs. illumination conditions

- Different illumination:

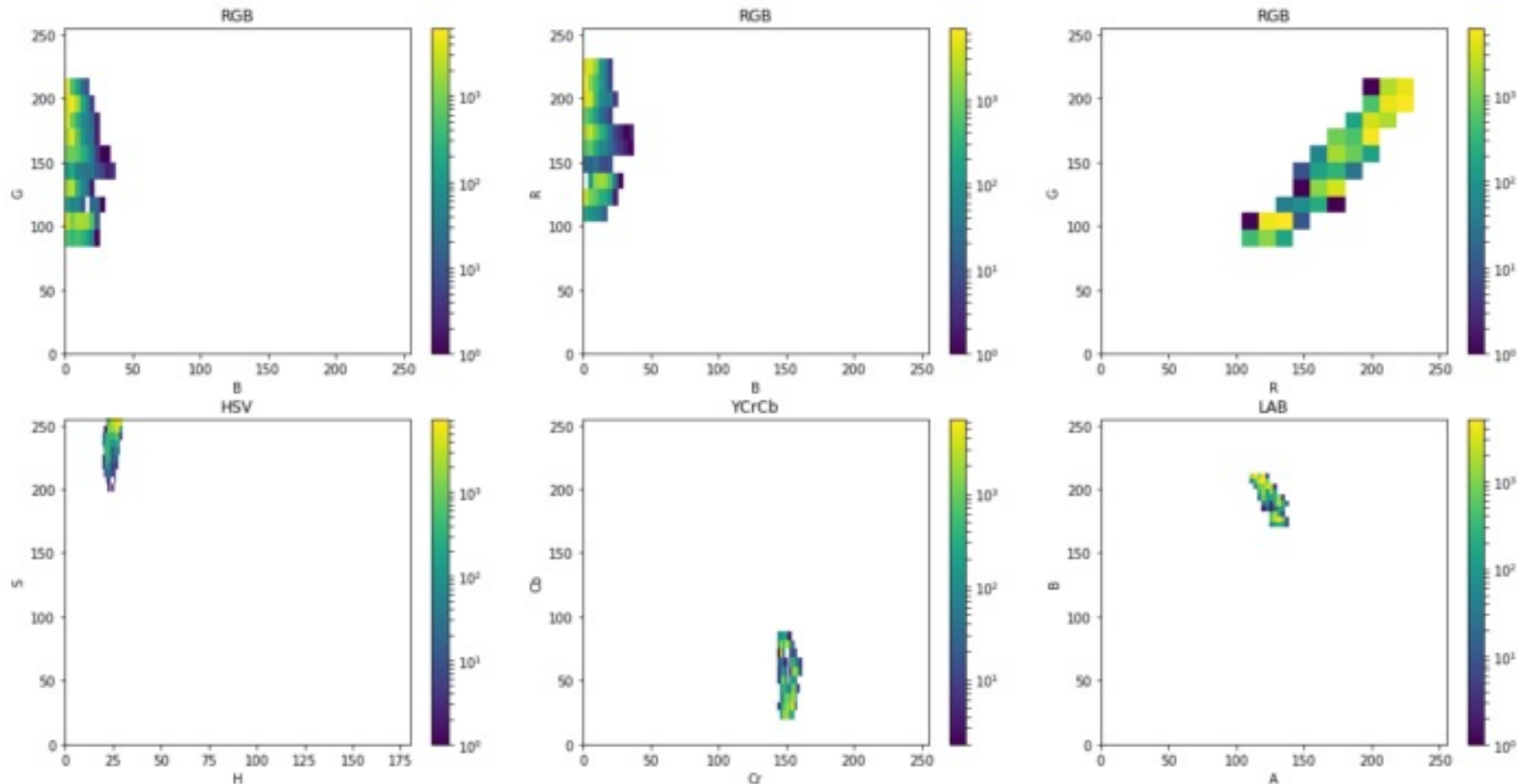


Fig.: Density Plot showing the variation of values in color channels under varying illumination for the **yellow color**

Source: Vikas Gupta, Learn OpenCV

# Color space vs illumination conditions

- Different illumination:
  - RGB space: the variation in the value of channels is very high
  - HSV: compact in **H**. Only H contains information about the absolute color → select
  - YCrCb, LAB: compact in **CrCb** and in **AB**
    - Higher level of compactness is in LAB
- Convert to other color spaces (OpenCV):
  - `cvtColor(bgr, ycb, COLOR_BGR2YCrCb);`
  - `cvtColor(bgr, hsv, COLOR_BGR2HSV);`
  - `cvtColor(bgr, lab, COLOR_BGR2Lab);`

# Image representation

## Continuous Images as functions

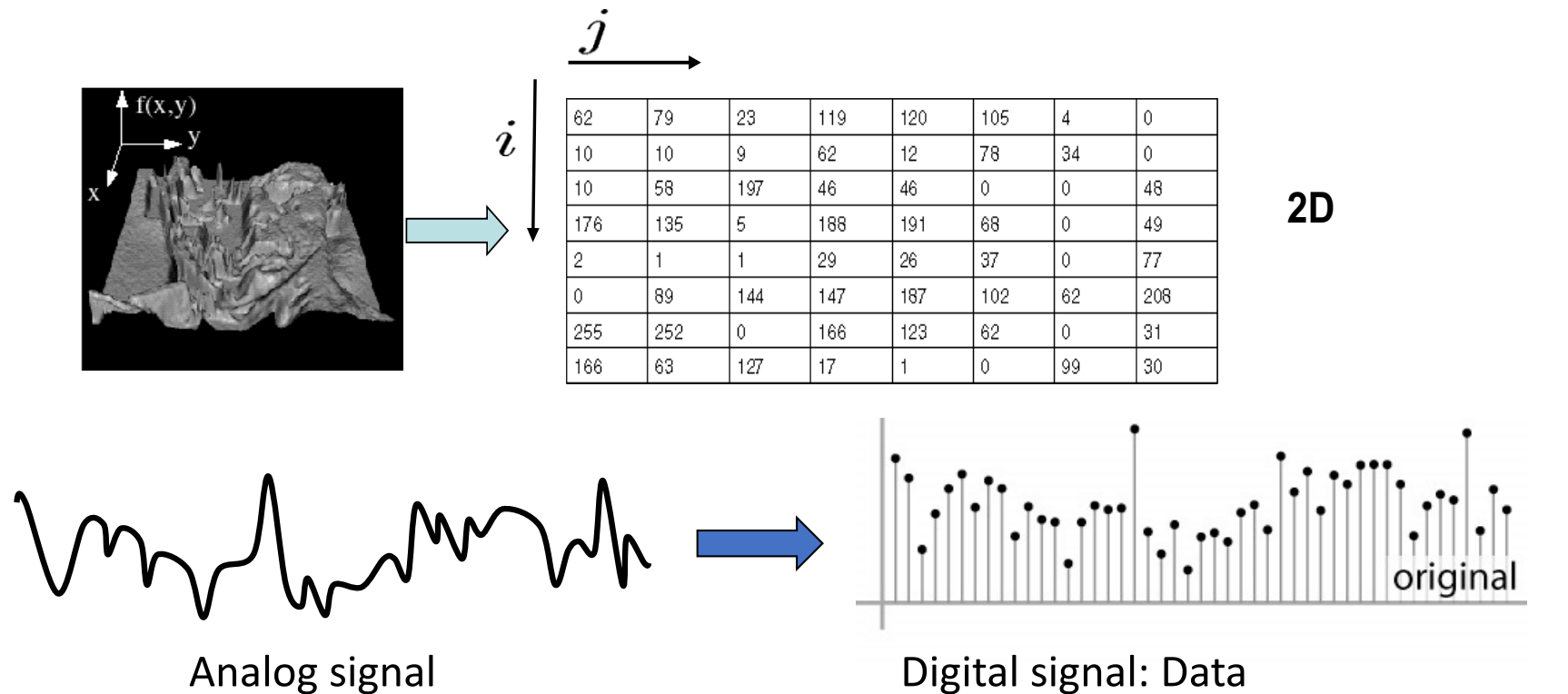
- Monochromatic Image: A continuous brightness function of a number of variables  $f$ , from  $\mathbb{R}^2$  to  $\mathbb{R}$ :
  - $f(x, y)$  gives the intensity at position  $(x, y)$
  - Realistically, we expect the image only to be defined over a rectangle, with a finite range
- A color image include 3 brightness functions of 3 color pasted together (3 color component signals). We can write this as a “vector-valued” function:

$$f(x, y) = \begin{bmatrix} r(x, y) \\ g(x, y) \\ b(x, y) \end{bmatrix}$$



# Digital images representation

- Sample the 2D space on a regular grid is pixel
- Quantize each sample (round to nearest integer)
- Image data is represented as a matrix of integer values.



Adapted from S. Seitz

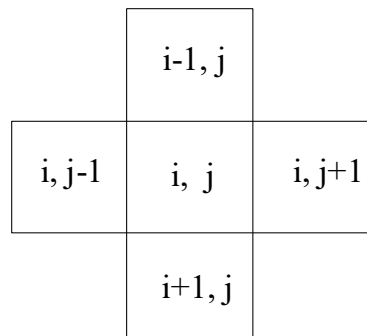
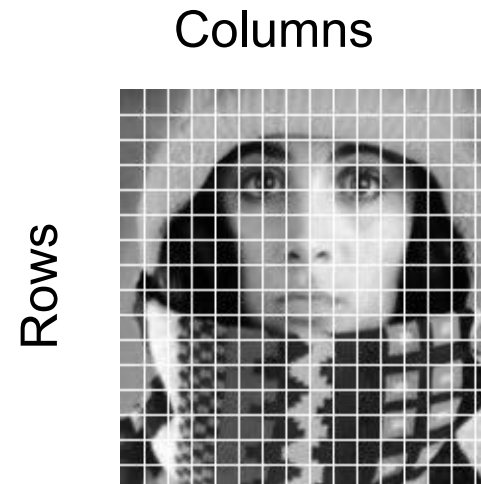
# Definition: Digital images

- Digital image functions  $f$  represented as matrices  $X(i,j)$ .
- **Image data** is represented by a rectangular array of integers
- An integer represents the brightness or darkness of the monochromatic image at that point (pixel). Limited brightness integer values (8 bit) = gray levels = values 0 to 255
- **Definition: Digital image is a matrix  $X(i,j)$  of pixels**, N:number of rows, M: number of columns, Q: integer brightness values (levels) of pixels

$$X(i, j) = \begin{matrix} & f(0,0) & f(0,1) & \dots & f(0,M-1) \\ f(1,0) & f(1,1) & \dots & f(1,M-1) \\ \dots & \dots & \dots & \dots \\ f(N-1,0) & f(N-1,1) & \dots & f(N-1,M-1) \end{matrix}$$

# Digital gray image

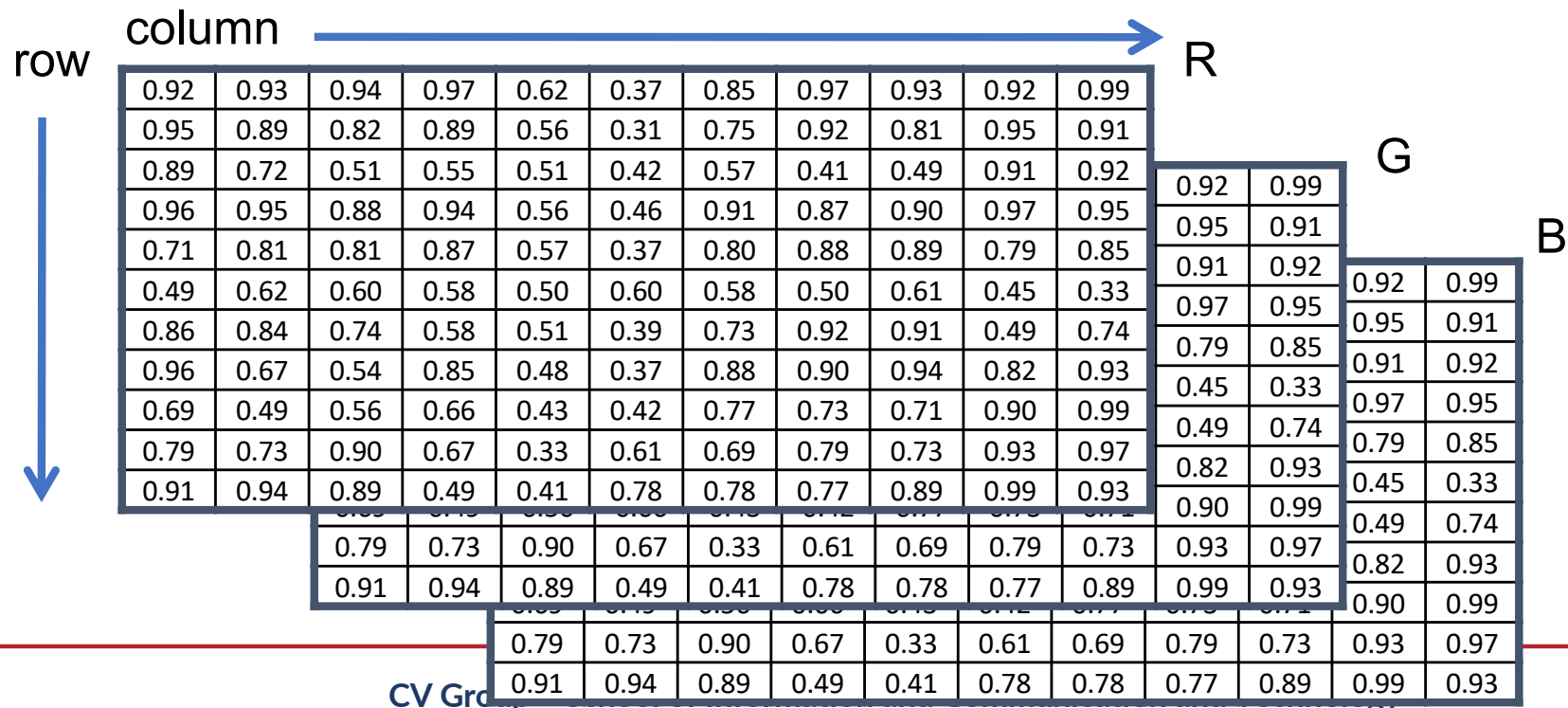
- Example: Matrix  $X(i,j)$  of pixels of a gray level image
- Image data: 2D array  $X(i,j)$  of integer brightness value uint8 of pixels at coordinates  $(i,j)$ .



|            |          |            |
|------------|----------|------------|
| $i-1, j-1$ | $i-1, j$ | $i-1, j+1$ |
| $i, j-1$   | $i, j$   | $i, j+1$   |
| $i+1, j-1$ | $i+1, j$ | $i+1, j+1$ |

# RGB color images in Matlab

- Images represented as a matrix  $X(i,j)$
- Suppose we have a  $N \times M$  RGB image called "Im"
  - $\text{Im}(1,1,1)$  = top-left pixel value in R-channel
  - $\text{Im}(y, x, b)$  = y pixels down, x pixels to right in the  $b^{\text{th}}$  channel
  - $\text{Im}(N, M, 3)$  = bottom-right pixel in B-channel
- `imread(filename)` returns a uint8 image (values 0 to 255)
  - Convert to double format (values 0 to 1) with `im2double`



## ❖ Parameters for digital image formats:

- **Digital image resolution:** (height x width) in pixels
- **Quantization** (bits per pixel):
  - Gray level image: 8 bits/ pixel
  - RGB color image: 24 bits/ pixel
  - Binary image: 1 bit/ pixel

## ❖ Digital Image Storage: file stored in two parts: Header; Data

## ❖ Common image file formats:

- GIF (Graphic Interchange Format) -
- PNG (Portable Network Graphics)
- JPEG (Joint Photographic Experts Group)
- TIFF (Tagged Image File Format)
- PGM (Portable Gray Map)
- FITS (Flexible Image Transport System)

# Digital video format

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- Parameters for digital video formats
  - Digital image resolution (height x width) in pixels
  - Quantization (bits per pixel)
  - Frame rate (frames per second)
- Standard video file formats
  - AVI, M-JPEG,
  - H26X (ITU\_T:H.261, H.263, H.263, H264)
  - MPEG-1, MPEG-2, MPEG-4 Part 10 / H264 AVC,mp4...



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**THANK YOU !**